

NANYANG TECHNOLOGICAL UNIVERSITY

School of Social Sciences

HE3002 Macroeconomics III

LECTURE 3

Balance of Payments and the Current Account

READINGS

1. Olivier Blanchard *Macroeconomics*, 8th Edition (2021), Prentice Hall, Chapters 17 and 18.

THE BALANCE OF PAYMENTS

1. A country's international transactions ^{Open economy} are recorded in the **balance-of-payments**, *BP*.

2. BP has three main components:

- (a) Current account, *CA*
- (b) Financial account, *FA*, and
- (c) Capital account, *KA*.

3. Therefore, *BP* is usually written as:

$$BP = CA + FA + \overset{O}{\underset{||}{KA}} \quad (1)$$

4. Since capital account balance is usually negligible, we are not discussing it here. Hence,

$$BP = CA + FA \quad (2)$$

5. The current account records (a) ^X exports and ^{IM} imports of goods and services and (b) international receipts or payments of income.

6. Exports and income receipts enter with a plus (+) and imports and income payments enter with a minus (-).

7. For example, if a US resident buys a smartphone from South Korea for \$500, then the US current account goes down by \$500. This is because this transaction represents an import of goods worth \$500. Accordingly, the US current account records a negative entry of \$500.

8. The financial account keeps record of transactions in financial assets between residents and non-residents. Sales of assets to nonresidents represent an export of an asset and are given a positive sign (+) (financial inflows) in the financial account. Purchases of assets from nonresidents represent an import of a financial asset and enter the financial account with a negative sign (-) (financial outflows).

Sales of assets to non-residents (foreigners) → Foreigners' holding of domestic assets → FI (+) 1 Liabilities

Purchases of assets by domestic residents → Domestic residents' holding of foreign assets → FO(-) Assets

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$$BP = CA + FA + KA \quad (1)$$

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7. For example, if a US resident buys a smartphone from South Korea for \$500, then the US current account goes down by \$500. This is because this transaction represents an import of goods worth \$500. Accordingly, the US current account records a negative entry of \$500.
US: (-) IM ↑ → (X-IM) ↓ → CA ↓ → BP ↓

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Sales of assets to non-residents (foreigners) → Foreigners' holding of domestic assets → FI (+) | Liabilities

Purchases of assets by domestic residents → Domestic residents' holding of foreign assets → FA (-) Assets

Note : $IM \uparrow \rightarrow (X-IM) \downarrow \rightarrow CA \downarrow \rightarrow BP \downarrow$
 $\text{Pay foreigners} \rightarrow FI \uparrow \rightarrow (FI-FO) \rightarrow FA \uparrow \rightarrow BP \uparrow \rightarrow \overline{BP}$

9. For example, in the case of the import of the smartphone, suppose the US resident pays for the phone with US currency, then this represents a sale (export) of US financial assets (currency) to a South Korean resident (Samsung) in the amount of \$500. Accordingly, the US financial account records a positive entry of \$500.

10. The smartphone example illustrates a fundamental principle of balance-of-payments accounting known as double-entry bookkeeping. $\rightarrow FO \quad FI (+)$

11. Each transaction enters the balance of payments twice, once with a positive sign and once with a negative sign. $(-)$

12. An implication of the double-entry bookkeeping methodology is that any change in the current account must be reflected in an equivalent change in the country's financial account, that is, the current account equals the difference between a country's purchases of assets from foreigners and its sales of assets to them, which is the financial account preceded by a minus sign. This relationship is known as the fundamental balance-of-payments identity.

13. Because any international transactions automatically gives rise to two offsetting entries in the BoP, we have $BP = CA + FA = 0 \rightarrow CA + (FI - FO) = 0 \rightarrow CA = -FA$

Current account balance = - Financial account balance

$$\therefore CA = (FO - FI) \quad (3)$$

14. The current account, on the other hand, consists of three components:

$$CA = TB + IB + NUT$$

Current account = Trade balance + Income balance + Net unilateral transfer (4)

$$CA = A - L \quad \text{owned}$$

$$CA > 0 \rightarrow A > L \rightarrow \text{Owned} > \text{Owed}$$

15. The trade balance or balance on goods and services is the difference between exports and imports of goods and services. It consists of

$$\text{China/Sing} \rightarrow \text{Creditor}$$

- (a) Merchandise trade balance or balance on goods, and

$$CA < 0 \rightarrow A < L \rightarrow \text{Owned} < \text{Owed}$$

- (b) Services balance.

$$US \rightarrow \text{Debtor}$$

16. The merchandise trade balance keeps record of net exports of goods.

17. The services balance keeps record of net receipts from items such as transportation, travel expenditures and legal assistance.

18. The income balance has

NII

- (a) Net investment income, and

- (b) Net investment compensation to employees.

Financial acc

① Portfolio (stock/share) - dividend

② other - Interest (IB)

NICE ③ Foreign direct investment (outward/inward) - Profit (IB)

19. Net investment income is the difference between income receipts on US-owned assets abroad and income payments on foreign-owned assets in the US. It includes items such as international interest, dividend payments and earnings of domestically owned firms operating abroad.

20. Net investment compensation to employees includes

- (a) As positive entries: (1) earnings of US residents employed temporarily abroad, (2) earnings of US residents employed by foreign governments in US, and (3) earnings of US residents employed by international organizations in US.
- (b) As negative entries: (1) foreign workers who commute to work in US, (2) foreign students studying in US, (3) foreign professionals temporarily residing in US, and (4) foreign temporary workers in US.

21. NUT Net unilateral transfers keeps record of the difference between gifts. One common item is private remittances.

22. The financial account has two main components as follows:

$$FA = FI - FO = L - A$$

Financial account = Increased in foreign-owned assets in the US (5)

– Increased in US-owned assets abroad (6)

(L)

23. Note that foreign-owned assets in the US includes FI ← Foreigners' holding of domestic assets

- ✓ (a) US securities held by foreign residents
- (b) US currency held by foreign residents
- ✓ (c) US borrowing from foreign banks
- ✓ (d) Foreign direct investment in the US

24. US-owned assets abroad are FO (A)

- ✓ (a) Foreign securities
- ✓ (b) US bank lending to foreigners
- ✓ (c) US direct investment abroad

25. An international transaction does not necessarily have to give rise to one entry in the current account and one entry in the financial account.

26. It can be the case that it gives rise to two offsetting entries in the financial account or two offsetting entries in the current account.

27. The example given earlier, namely, importing a smartphone and paying for it with cash, gives rise to one entry in the current account CA and one entry in the financial account FA.

28. International transactions that involve the exchange of financial assets generate two entries in the financial account and no entry in the current account. For example, if a US resident purchases FA (+) & (-) BP = 0

Buy Italian shares (FO) (-)

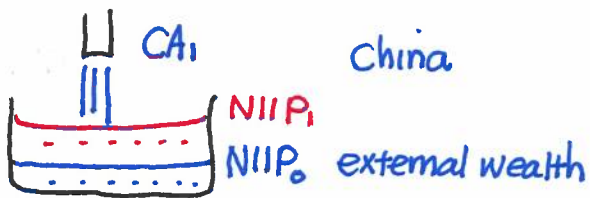
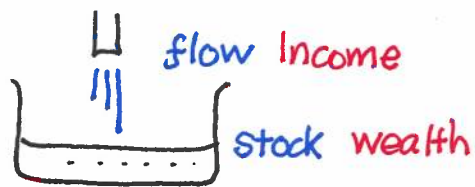
Pay US\$ → Foreigners' holding of domestic assets (US\$) (FI) (+)
shares from Italy paying with dollars, then the financial account receives both a positive entry (the sale, or export, of dollars to Italy) and a negative entry (the purchase, or import, of equity shares from Italy).

29. It can also be the case that a transaction generates two offsetting entries in the current account and no entry in the financial account.
30. Suppose the US donates medications worth \$10 million to an African country afflicted by Malaria. This gift gives rise to two entries in the current account.
- (a) The export of the malaria medication would be recorded in the merchandise trade balance, which is part of the current account, with +\$10 million. $X (+) \rightarrow CA \uparrow \rightarrow BP \uparrow$
 - (b) As the US does not receive an item of economic value in return, the offset is an entry of -\$10 million in net unilateral transfers, which is also part of the current account. $NUT (-) \rightarrow CA \downarrow \rightarrow BP \downarrow$
31. There is another type of transaction that merits separate discussion. This type of transaction is the purchase or sale of official reserve assets by central banks.
32. Official international reserves are foreign assets held by central banks as a cushion against national economic misfortune. At one time, official reserves consisted largely gold, but today central bank's reserves include substantial foreign financial assets.
33. Central banks often buy or sell international reserves in private asset markets to affect macroeconomic conditions in their economies. Official transactions of this type are called official foreign exchange interventions.

[Table 1: The US balance of payments in 2018 in billions of USD]

Singapore

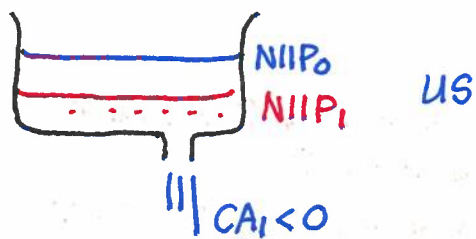
$X \uparrow \rightarrow$ To pay for this, foreigners buy S\$ $\rightarrow D_{S\$} \uparrow \rightarrow P_{S\$} \uparrow$
 $\rightarrow S\$ \text{ app} \rightarrow$ To prevent app $\rightarrow$ sell S\$ $\rightarrow M_3 \uparrow$
 $\rightarrow$ buy US\$ $\rightarrow$ foreign reserves $\uparrow$



$$NIIP_0 + CA_1 = NIIP_1$$

$$\therefore CA_1 = NIIP_1 - NIIP_0 = \Delta NIIP$$

$$CA_1 > 0 \rightarrow \Delta NIIP > 0 \rightarrow NIIP_1 > NIIP_0 \rightarrow NIIP \uparrow$$



$$CA_1 < 0 \rightarrow \Delta NIIP < 0 \rightarrow NIIP_1 < NIIP_0 \rightarrow NIIP \downarrow$$

flow
CA

Stock
NIIP

THE CURRENT ACCOUNT AND THE NET INTERNATIONAL INVESTMENT POSITION

1. One reason why the concept of current account balance is economically important is that it reflects a country's net borrowing needs. For example, in 2018 the US ran a current account deficit of \$488.5 billion. To pay for this deficit, the country must have either
 - (a) reduced part of its international asset position or
 - (b) increased its international liability position or
 - (c) both.
$$CA = -FA$$
$$CA = -(FI - FO)$$
$$CA = -(L - A) \Rightarrow CA = A - L$$
2. In this way, the current account is related to changes in a country's net international investment position (NIIP).
3. The term net international investment position is used to refer to a country's net foreign wealth, that is, the difference between the value of foreign assets owned by the country's residents and the value of the country's assets owned by foreign residents.
4. The net international investment position is a stock, while the current account is a flow.

[Figure 1: The US current account and net international investment position]

5. The US net international investment position was positive at the beginning of the sample. However, in the early 1980s, the US began running large current account deficits.
6. By 1989 these deficits had eroded the net foreign wealth of the US and she became a net debtor to the rest of the world for the first time since World War I.
7. The US current account deficits of the 1980s did not turn out to be temporary. As a consequence, by the end of the 1990s, the US had become the world's largest external debtor.
8. Moreover, the current account deficits continued to rise for twenty five years. Only shortly before the onset of the Great Recession of 2008, did the downward trend stop and current account deficits stabilized. In fact, they became smaller in magnitude.
9. By the end of 2018, the net international investment position of the US stood at -9.6 trillion dollars or -47 percent of GDP. This is a big number, and many economist wonder whether the observed downward trend in the net international investment position will be sustainable over time.
10. The concern stems from the fact that countries that accumulated large external debt to GDP ratios in the past, such as many Latin American countries in the 1980s, Southeast Asian countries in the 1990s, and more recently peripheral European countries, have experienced sudden reversals in international capital flows that were followed by costly financial and economic crises.

financial

SAVING, INVESTMENT, AND THE CURRENT ACCOUNT BALANCE

- Recall that the equilibrium in the goods market suggests that investment equals to saving, which is the sum of private saving and public saving.

- We can now derive the corresponding condition for the open economy.

- At equilibrium,

$$Y = C + I + G + (X - \frac{IM}{\epsilon}) \quad (7)$$

- Rearrange gives:

$$Y - T - C = I + (G - T) + NX \quad (8)$$

where $NX = X - IM/\epsilon$.

- Recall that, in an open economy, the total income of domestic residents is equal to domestic income, which is equal to domestic output, Y , plus net income from abroad, NI .

- Add NI to both sides of the equation gives:

$$(Y + NI - T) - C = I + (G - T) + (NX + NI) \quad (9)$$

where $(Y + NI - T) - C$ is the private saving, S_p and $T - G$ is the public saving, S_g and $NX + NI$ is the current account, CA .

- As a result,

$$S_p = I + (G - T) + CA \quad (10)$$

Budget surplus = $T - G > 0$
 $S_g > 0$

- Or,

$$CA = S_p + (T - G) - I \quad (11)$$

- And

$$CA = S_p + S_g - I \quad (12)$$

- Given national saving, $S = S_p + S_g$, then:

$$CA = S - I \quad (13)$$

$CA > 0 \rightarrow S > I$ China / S_g
 $CA < 0 \rightarrow S < I$ US

- The current account balance is equal to national saving (private plus public) minus investment.

(a) A current account **surplus** implies that the country is saving **more than** it invests.

(b) A current account **deficit** implies that the country is saving **less than** it invests.

- Note some of the things Equation (13) says:

- (a) An increase in investment must be reflected in either an increase in private or public saving or a deterioration in CA - a smaller CA surplus, or a larger CA deficit, depending on whether the current account is initially in surplus or deficit.
- (b) A deterioration in the government budget balance - either a smaller budget surplus or a larger budget deficit - must be reflected either in an increase in private saving or a decrease in investment, or a deterioration of the CA.
- (c) A country with a higher saving rate (private plus public) must have either a high investment or a large CA surplus.

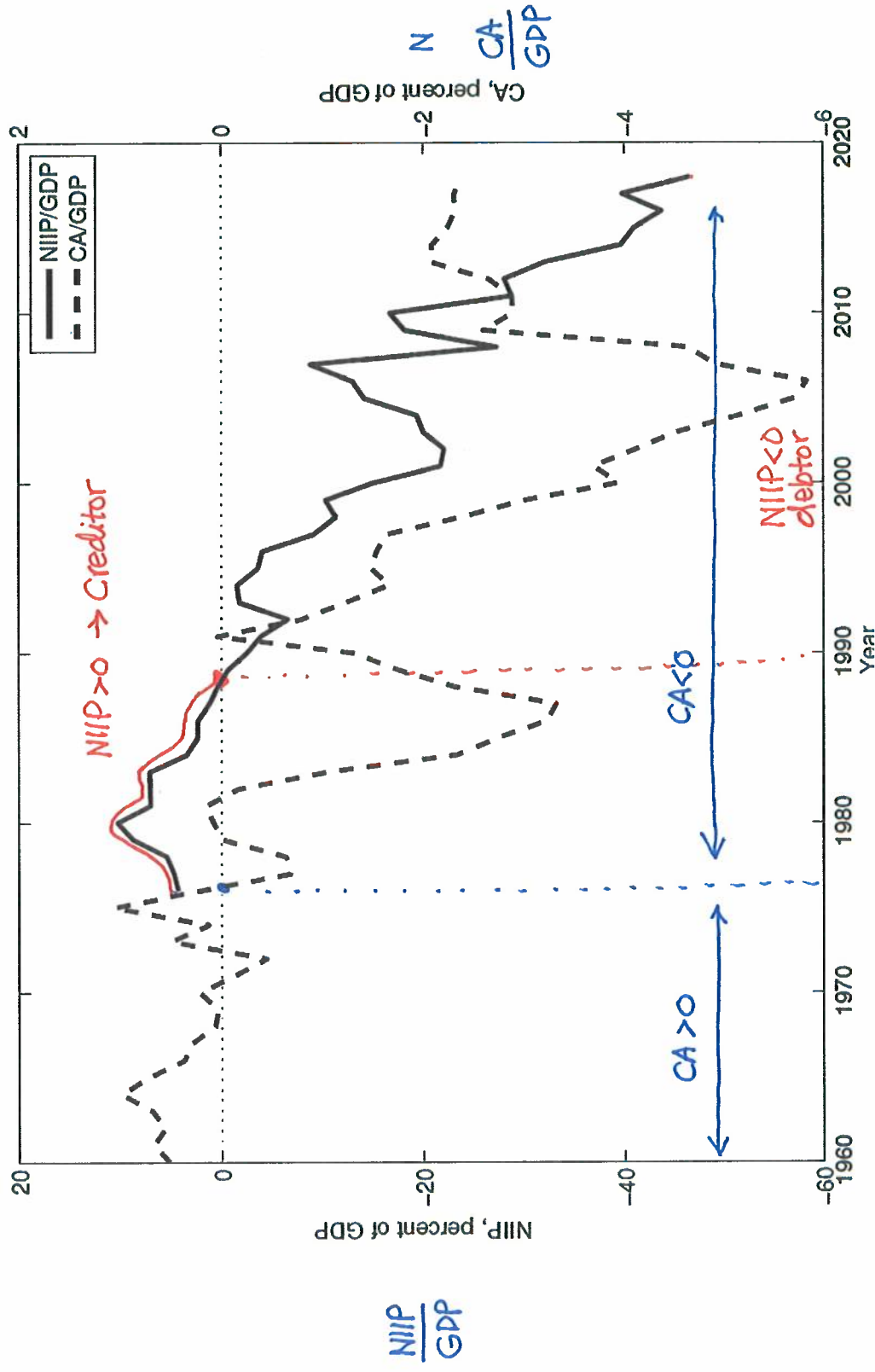
WHO LENDS AND WHO BORROWS AROUND THE WORLD?

1. The large observed US current account deficits must be matched by current account surpluses of other countries with the US.
2. Over the past decade, an increasing fraction of the US current account deficit is accounted for by current account deficits with China.

[Figure 2: Cumulative Current Account Balances Around the World: 1980-2017]

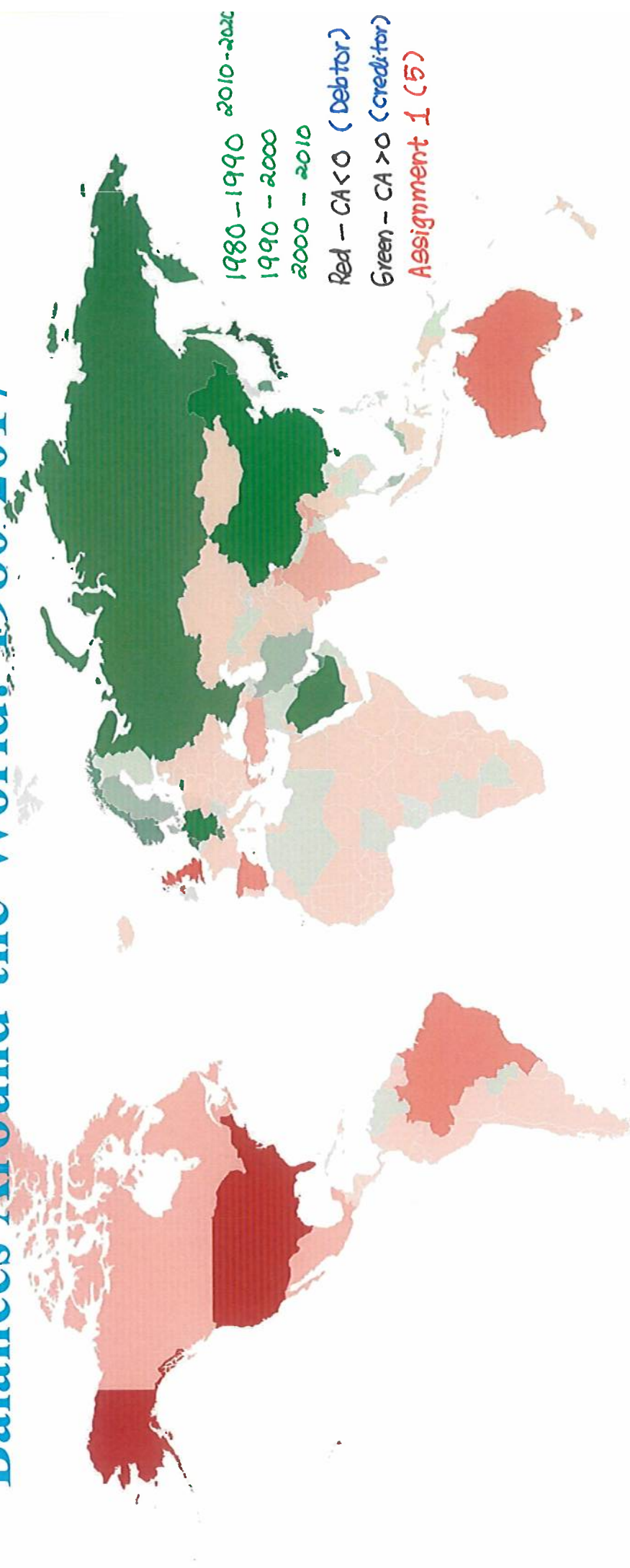
3. The US current account with China as a fraction of the total US current account balance was about 20% in 1999 and has been increasing steadily, reaching a peak of 70% in 2009.

Figure 1: The US Current Account and Net International Investment Position



Notes. CA, NIIP, and GDP stand for current account, net international investment position, and gross domestic product, respectively. The sample period for CA is 1960 to 2018 and for NIIP 1976 to 2018. The data source is <http://www.bea.gov>.

Figure 2: Cumulative Current Account Balances Around the World: 1980-2017



Note. The map shows for each country the sum of current account balances (in billions of U.S. dollars) between 1980 and 2017. Cumulative current account surpluses appear in green and cumulative deficits in red. There are six shades of red and green corresponding, respectively, at least one half, one fourth, one eighth, one sixteenth, and one thirty-second of the maximum cumulative current account deficit (U.S.: $-\$11,035bn$) and the maximum cumulative current account surplus (Japan: $\$3,906bn$). The data source is Philip R. Lane and Gian Maria Milesi-Ferretti (2017), "International Financial Integration in the Aftermath of the Global Financial Crisis," IMF Working Paper 17/115. Data for former Soviet Union countries start in 1992. Countries for which no data are available appear in gray. Country names are displayed for the countries with the top 10 largest cumulated current account surpluses and deficits.

LECTURE 4
Output, the Interest Rate, and the Exchange Rate

READINGS

- Olivier Blanchard *Macroeconomics*, 8th Edition (2021), Prentice Hall, Chapter 19.

INTRODUCTION

- In some countries, exchange rate is not a policy instrument. Rather, it is determined in the foreign exchange market.
flexible exchange regime
Japan & US
 - What determines the exchange rate? *flexible*
 - How can policymakers affect it? *MP/FP*
- To answer these questions, we reintroduce financial markets and examine the implications of equilibrium in both the goods market and financial markets, including the foreign exchange market.
IS LM WIP
- This allows us to characterize the joint movements of output, the interest rate, and the exchange rate in an open economy.
Y i ← Nominal interest rate ✓
r ← real interest rate
- This is an extension of the IS-LM model and is known as the Mundell-Fleming model.
E open

EQUILIBRIUM IN THE GOODS MARKET

- Recall that equilibrium in the goods market gives:
r ↑ → cost of borrowing ↑ → I ↓
ε ↑ → real app → X expensive → X ↓
M cheaper → M ↑
X-M
NX ↓
$$Y = C + I + G + (X - M) \quad \text{exo} \quad \begin{matrix} + & - & + & + \\ + & - & + & + \end{matrix}$$
$$Y = C(Y - T) + I(Y, r) + G + X(Y^*, \epsilon) - \frac{IM(Y, \epsilon)}{\epsilon} \quad (1)$$

- For the goods market to be in equilibrium, output (LHS) must be equal to the demand for domestic goods (RHS).

- Since

$$NX(Y, Y^*, \epsilon) = X(Y^*, \epsilon) - \frac{IM(Y, \epsilon)}{\epsilon} \quad (2)$$

then,

$$Y = C(Y - T) + I(Y, r) + G + NX(Y, Y^*, \epsilon) \quad (3)$$

As $\epsilon = E \overset{1}{P/P^*}$, and we let $P/P^* = 1$, then, $\epsilon = E$. Therefore, we have at goods market equilibrium:

$$Y = C(Y - T) + I(Y, r) + G + NX(Y, Y^*, E) \quad (4)$$

EQUILIBRIUM IN FINANCIAL MARKETS

1. When we looked at financial markets in the IS-LM model, we assumed that people choose between two financial assets: money and bonds. *closed*
2. Now that we look at a financially open economy, we must take into consideration that they can choose between domestic bonds and foreign bonds. *open*

Domestic Bonds vs Foreign Bonds

1. As we look at the choice between domestic bonds and foreign bonds, we assume that financial investors, domestic or foreign, go for the highest expected rate of return.
2. This implies that, in equilibrium, both domestic bonds and foreign bonds must have the same expected rate of return; otherwise, investors would be willing to hold only one or the other, but not both, and this could not be an equilibrium.
3. This assumption implies that the following arbitrage relation – the interest parity condition – must hold:

$$(1 + i_t) = (1 + i_t^*) \left(\frac{E_t}{E_{t+1}^e} \right) \quad (5)$$

where i_t is the domestic interest rate, i_t^* is the foreign interest rate, E_t is the current exchange rate, and E_{t+1}^e is the expected future exchange rate. *Note: if ceteris paribus, $i \uparrow \rightarrow i > i^* \rightarrow$*

4. Reorganize this gives:

$$E_t = \frac{1 + i_t}{1 + i_t^*} E_{t+1}^e \quad (6)$$

5. When E_{t+1}^e is taken as given, then we have:

$$E = \frac{1 + i}{1 + i^*} \bar{E}^e \quad (7)$$

6. This relation tells us that current exchange rate depends on the domestic interest rate, the foreign interest rate, and the expected future exchange rate.

- (a) An increase in the domestic interest rate leads to an increase in the exchange rate. *$i \uparrow \rightarrow E \uparrow$*
- (b) An increase in the ~~domestic~~ *foreign* interest rate leads to a decrease in the exchange rate. *$i^* \uparrow \rightarrow E \downarrow$*

$$\bar{E}^e \uparrow \rightarrow E \uparrow$$

(c) An increase in the expected future exchange rate leads to an increase in the current exchange rate.

7. Figure 1 shows the relation between the interest rate and the exchange rate implied by interest parity.

[Figure 1: Interest parity relation]

8. The relation is drawn for a given expected future exchange rate, $\bar{E}^e$, and a given foreign interest rate, i^* , and is represented by an upward-sloping line: the higher the domestic interest rate, the higher the exchange rate.
9. When the domestic interest rate is equal to the foreign interest rate ($i = i^*$), the exchange rate is equal to the expected future exchange rate ($E = \bar{E}^e$).
10. This implies that the line corresponding to the interest parity condition goes through point 0 in Figure 1.
11. This relation suggests that an increase in domestic interest rate relative to foreign interest rate leads to an appreciation of domestic currency.

$$E = \frac{1+i}{1+i^*} \cdot \bar{E}^e$$



When $i \uparrow \rightarrow E \uparrow$
UIP upward

PUTTING GOODS AND FINANCIAL MARKETS TOGETHER

1. We now have the elements we need to understand the movements of output, the interest rate, and the exchange rate. E
2. Goods market equilibrium implies that output depends, among other factors, on the interest rate and the exchange rate:

Goods mkt $\rightarrow$ $(Y) = C(Y - T) + I(Y, r) + G + NX(Y, Y^*, E)$ (8)

IS $\frac{1+i}{1+i^*} \cdot \bar{E}^e$

3. Now, treat the interest rate, i as the policy rate set by the central bank:

Financial / money mkt $i = \bar{i}$ (9)

LM

4. And the interest parity condition implies a positive relation between the domestic interest rate and the exchange rate:

UIP $E = \frac{1+i}{1+i^*} \bar{E}^e$ (10)

5. Together, these three relations determine output, the interest rate and the exchange rate.
6. Working with three relations is not very easy, but we can easily reduce them to two by using the interest parity condition to eliminate the exchange rate in the goods market equilibrium relation.
7. Doing this gives us the following two equations, the open-economy versions of our familiar IS and LM relations:

$Y = C(Y - T) + I(Y, r) + G + NX(Y, Y^*, \frac{1+i}{1+i^*} \bar{E}^e)$ (11)

$i = \bar{i}$ (12)

8. Together, the two equations determine the interest rate and equilibrium output.
9. Then, using Equation (10) we get the implied exchange rate.
10. Take the IS relation first and consider the effects of an increase in the interest rate on output. An increase in the interest rate now has two effects:

- $i \uparrow \rightarrow p$ is fixed, i & r move together $\rightarrow r \uparrow \rightarrow$ cost of borrowing $\uparrow \rightarrow I \downarrow \rightarrow ZZ \downarrow$
- (a) The first effect, which was already present in a closed economy, is the direct effect on investment: a higher interest rate leads to a decrease in investment, a decrease in the demand for domestic goods and a decrease in output. $\rightarrow Y = Z$
 $\rightarrow Y \downarrow$
- IS downward.

- (b) The second effect, which is present only in the open economy, is the effect through the exchange rate: an increase in the domestic interest rate leads to an increase in the exchange rate
- $i \uparrow \rightarrow i > i^* \rightarrow$ Domestic bond is attractive $\rightarrow$ buy domestic bonds $\rightarrow$ D domestic currency $\uparrow \rightarrow E \uparrow$
- $\rightarrow X$ expensive $\rightarrow X \downarrow$ $\rightarrow M$ cheap $\rightarrow M \uparrow$ $\rightarrow NX = X - M$
- $ZZ \downarrow \rightarrow Y = Z (Y \downarrow)$ IS downward

rate – an appreciation. The appreciation, which makes domestic goods more expensive relative to foreign goods, leads to a decrease in net exports and, therefore, to a decrease in the demand for domestic goods and a decrease in output.

11. Both effects work in the same direction: an increase in the interest rate decreases demand directly and indirectly – through the adverse effect of the appreciation on demand.
12. As an increase in the interest rate reduces output, the IS curve is downward-sloping.
13. The LM curve is the same as in the closed economy: it is a horizontal line, at the level of the interest rate $\bar{i}$ set by the central bank.

[Figure 2: The IS-LM model in an open economy]

14. Equilibrium in the goods and financial markets is attained at point 0 with output level $Y = Y_0$ and interest rate $i = i_0 = \bar{i}$.
15. The equilibrium value of the exchange rate is associated with the equilibrium interest rate, $i = i_0 = \bar{i}$, is equal to $E = E_0 = \bar{E}^e$.
16. In summary,
 - (a) The IS curve is downward-sloping – An increase in the interest rate leads directly and indirectly (through the exchange rate) to a decrease in demand and a decrease in output.
 - (b) The LM curve is horizontal at the interest rate set by the central bank.
17. Equilibrium output and the equilibrium interest rate are given by the intersection of the IS and the LM curves. Given the foreign interest rate and the expected future exchange rate, the equilibrium interest rate determines the equilibrium exchange rate.

available want

THE EFFECTS OF POLICY IN AN OPEN ECONOMY

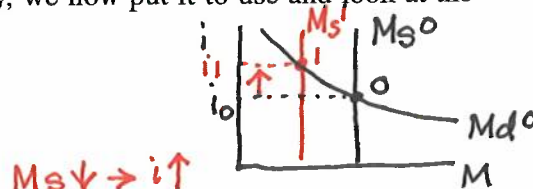
Note: $M_s \downarrow \rightarrow M_s < M_d \rightarrow \text{sell bonds} \rightarrow P_b \downarrow \rightarrow i \uparrow$

1. Having derived the IS-LM model for the open economy, we now put it to use and look at the effects of policy.

Case 1:

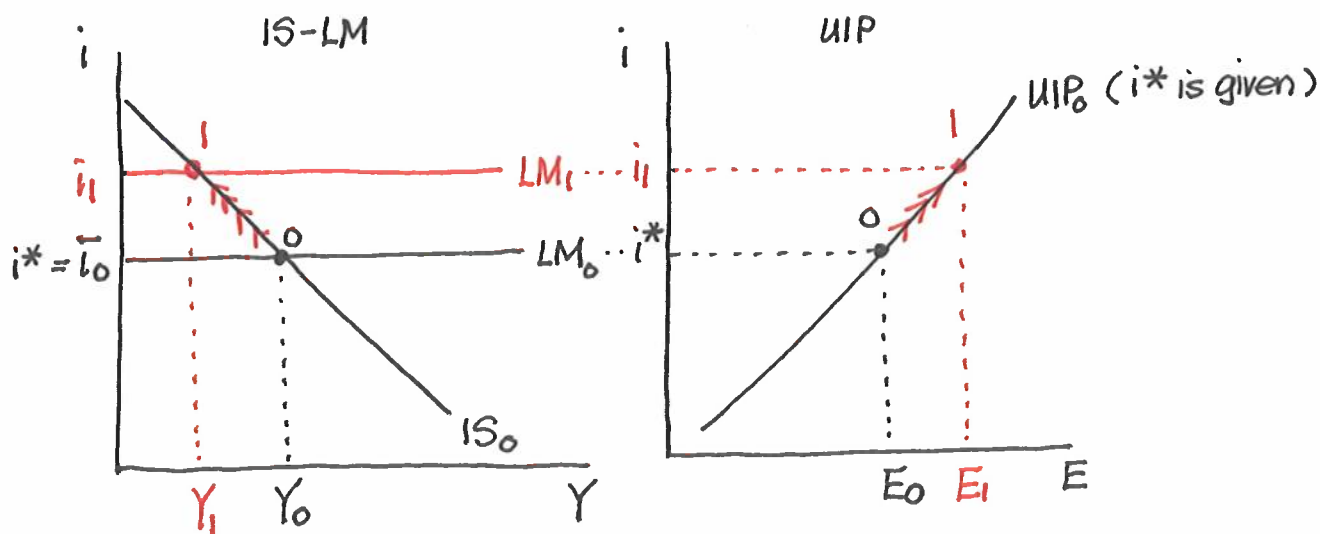
The Effects of Monetary Policy in an Open Economy

1. At a given level of output, a decrease in the money stock by, say, $\Delta M < 0$ leads to an increase in the interest rate: the LM curve shifts up, from LM_0 to LM_1 . Because money does not directly enter the IS relation, the IS curve does not shift.



[Figure 3: The effect of an increase in the domestic interest rate]

2. The equilibrium moves from point 0 to point 1. The increase in the interest rate leads to an appreciation.
3. So, in an open economy, monetary policy works through two channels:
 - (a) As in the closed economy, it works through the effect of the interest rate on spending.
 - (b) It works through the effect of the interest rate on exchange rate and the effect of the exchange rate on exports and imports.
4. Both effects work in the same direction.
5. A monetary contraction leads to an increase in the interest rate, making domestic bonds more attractive and triggering an appreciation.
6. The higher interest rate and the appreciation both decrease demand and output.



Initially $Y = Y_0$, $i = i_0$, $E = E_0$ at pt 0.

Suppose now $M_s \downarrow \rightarrow i \uparrow (i_0 \rightarrow i_1) \rightarrow i > i^* \rightarrow \text{Domestic bond is attractive} \rightarrow$
 $\dots \rightarrow E \uparrow (E_0 \rightarrow E_1) \rightarrow \begin{cases} \text{exp} \downarrow \rightarrow X \downarrow \\ \text{M cheap} \rightarrow M \uparrow \end{cases} \rightarrow NX = X - M \rightarrow NX \downarrow \rightarrow Z \downarrow \rightarrow Y = Z$
 $Y \downarrow$
 $(Y_0 \rightarrow Y_1)$

Case 2:

The Effects of Fiscal Policy in an Open Economy with an Unchanged Interest Rate

MP

$G \uparrow$ and/or $T \downarrow$

- Now that we have looked at monetary policy, we look at an expansionary fiscal policy.
- Suppose that, starting from a balanced budget, the government decides to increase defence spending without raising taxes and so runs a budget deficit. $T = G$ $G \uparrow (G_0 \rightarrow G_1)$ $\downarrow$ $\text{Budget} = \bar{T} - G \rightarrow \text{Budget} < 0$
- The economy is initially at point 0 with $Y = Y_0$ and $i = i_0 = \bar{i}$. The increase in government spending by, say, $\Delta G > 0$ increases output at a given interest rate, shifting the IS curve to the right, from IS_0 to IS_1 .
- Because the central bank does not change the policy rate, the LM curve does not shift. $\bar{MP}$

[Figure 4: The effects of an increase in government with unchanged interest rate]

- The new equilibrium is at point 1, with a higher level of output at Y_1 .
- Because the interest rate has not changed, neither has the exchange rate.
- So, an increase in government spending, when the central bank keeps the interest rate unchanged, leads to an increase in output with no change in exchange rate. $Y_0 \rightarrow Y_1$ $E = E_0$ $Y = C + I + G + NX$ $Y \uparrow \rightarrow C = C(Y - T), C \uparrow$
- What happens to the various components of demand?

(a) Consumption and government spending increase: Consumption goes up because of the increase in income and government spending goes up because of the policy.

(b) Investment also rises because $I = I(Y, r)$. $\bar{r} \rightarrow \bar{r}$ but $Y \uparrow \rightarrow I = I(Y, r), I \uparrow$

(c) Since $NX = NX(Y, \bar{Y}^*, \bar{E})$, with $\bar{Y}^*$ and $\bar{E}$ unchanged, a higher output leads to an increase in imports and hence a decrease in NX .

$Y \uparrow (Y_0 \rightarrow Y_1) \rightarrow M \uparrow$ b/c $M = M(\bar{Y}, \bar{E}) \rightarrow NX = \bar{X} - M$ $\bar{X}$ b/c $X = X(\bar{Y}^*)$

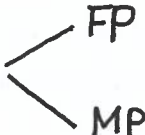
Case 3: FP+MP

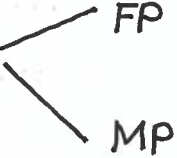
The Effects of Fiscal Policy in an Open Economy with the Central Bank Raising Interest Rate

1. Suppose $Y = Y_n$ but government needs to $G \uparrow$ $G_0 \rightarrow G_1$ increase spending to pay for an exceptional event, such as a big flood.
2. In this case, the central bank will worry that the increase in G may move the economy above Y_n . As a result, the central bank responds by an increase in the interest rate.
3. In this case, when IS shifts to the right, LM shifts up.

[Figure 5: The effects of an increase in government with higher interest rate]

4. The new equilibrium is at point 2 with a higher level of output and a higher interest rate. $Y_n \rightarrow Y_2$ $i_0 \rightarrow i_1$
5. The higher interest rate leads to an increase in the exchange rate – an appreciation. So an increase in government spending leads to an increase in output, an increase in the interest rate and an appreciation. $E_0 \rightarrow E_2$
 $Y \uparrow \rightarrow i \uparrow \& E \uparrow$ $I = I(Y, r)$
6. Can we tell what happens to the various components of demand? $Y \uparrow \rightarrow C \uparrow$ $G \uparrow$ (Policy) $i \uparrow \rightarrow r \uparrow \rightarrow I \downarrow$ $] \rightarrow I?$
 - (a) Consumption and government spending both go up. Consumption goes up because $C = C(Y - T)$, and government spending goes up by assumption.
 - (b) What happens to investment is ambiguous. Recall that investment depends on both output and the interest rate: $I = I(Y, i)$. On the one hand, output goes up, leading to an increase in investment. But on the other, the interest rate also goes up, leading to a decrease in investment. Depending on which of these two effects dominates, investment can go up or down.
 - (c) Recall that $NX = NX(Y, Y^*, E)$. Thus, both the increase in output and the appreciation combine to decrease net exports: the increase in output increases imports, and the appreciation decreases exports and increases imports. $Y \uparrow$
7. This version of IS-LM model for an open economy was first put together in the 1960s by Robert Mundell of Columbia and Marcus Fleming of the IMF.

Flexible Exchange  FP
MP

Fixed Exchange  FP
MP

FIXED EXCHANGE RATES

HK vs Singapore → Managed float

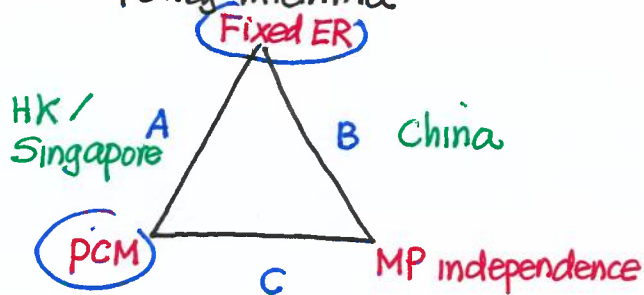
1. We have assumed so far that central bank chose the interest rate and let exchange rate adjust freely.
2. In many countries, central banks may go for different exchange rate regimes.

Pegs, Crawling Pegs, Bands, the EMS and the Euro

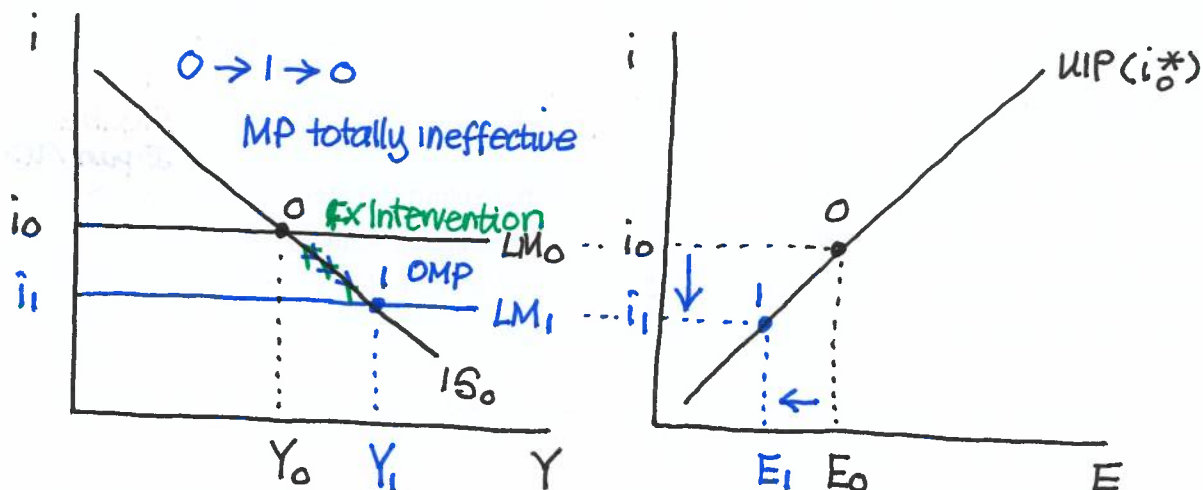


1. At one end of the spectrum are countries with **flexible exchange rates**, such as the USA or Japan. These countries do not have explicit exchange rate targets. Although their central banks probably do not ignore movements in the exchange rate, they have shown themselves to be quite willing to let their exchange rates fluctuate considerably.
2. At the other end are countries that operate under **fixed exchange rates**. These countries maintain a fixed exchange rate in terms of some foreign currency. Some peg their currency to the dollar. For example, from 1991–2001, Argentina pegged its currency, the **peso**, at the highly symbolic exchange rate of 1 dollar for 1 peso. Other countries used to peg their currency to the French franc (most of these are former French colonies in Africa); as the French franc has been replaced by the Euro, they are now pegged to the Euro.
3. Still other countries peg their currency to a basket of foreign currencies, with the weights reflecting the composition of their trade. For example, **Singapore**.
4. The label 'fixed' is a bit misleading: it is not the case that the exchange rate in countries with fixed exchange rates never actually changes. But changes are rare. Because these changes are rare, economists use specific words to distinguish them from the daily changes that occur under flexible exchange rates.
5. A decrease in the exchange rate under a regime of fixed exchange rates is called a **devaluation** rather than a **depreciation**, and an increase in the exchange rate under a regime of fixed exchange rates is called a **revaluation** rather than an **appreciation**.
6. Between these extremes are countries with various degrees of commitment to an exchange rate target. For example, some countries operate under a **crawling peg**. **Singapore → BBC**
7. The name describes it well: these countries typically have inflation rates that exceed the US inflation rate. If they were to peg their nominal exchange rate against the dollar, the more rapid increase in their domestic price level above the US price level would lead to a steady real appreciation and rapidly make their goods uncompetitive. To avoid this effect, these countries choose a predetermined rate of depreciation against the dollar. They choose to 'crawl' (move slowly) vis-à-vis the dollar.

Policy trilemma



Fixed Exchange + PCM + MP



Initially $LM_0 = IS_0$ at pt $O, Y = Y_0, i = i_0 = i_0^* = i^*, E = E_0$

Suppose now $i \downarrow (i_0 \rightarrow i_1)$. $i < i^* \rightarrow$ foreign bonds are more attractive $\rightarrow$ Buy foreign bonds $\rightarrow$ Domestic currency $\downarrow$
 $E \downarrow$

Under fixed exchange rate,

To prevent depⁿ $\rightarrow$ buy domestic currency $\rightarrow$ $(M_s \downarrow)$
 $\rightarrow$ sell foreign currency $\rightarrow$ Foreign reserve $\downarrow$

Before intervention

Owned Assets

Bonds

Foreign reserves

Owed Liabilities

Central bank money

After intervention (OMP + FX Interⁿ)

A

Bonds +100b

FR -100b

L

Money +100b -100b

8. Yet another arrangement is for a group of countries to maintain their bilateral exchange rates (the exchange rate between each pair of countries) within some bands. Perhaps the most prominent example was the European Monetary System (EMS), which determined the movements of exchange rates within the EU from 1978–1998. Under the EMS rules, member countries agreed to maintain their exchange rate vis-à-vis the other currencies in the system within narrow limits, or bands, around a central parity – a given value for the exchange rate. Changes in the central parity and devaluations or revaluations of specific currencies could occur, but only by common agreement among member countries.
9. After a major crisis in 1992, which led a number of countries to drop out of the EMS altogether, exchange rate adjustments became more and more infrequent, leading a number of countries to move one step further and adopt a common currency, the euro. The conversion from domestic currencies to the euro began on 1 January 1999, and was completed in early 2002.

① Monetary Policy When the Exchange Rate is Fixed

1. Suppose a country decides to peg its exchange rate at some chosen value, call it $\bar{E}$.
2. How does it actually achieve this? The government cannot just announce the value of the exchange rate and stand there. Rather, it must take measures so that its chosen exchange rate will prevail in the foreign exchange market. Let's look at the implications and mechanics of pegging.
3. Pegging or no pegging, the exchange rate and the nominal interest rate must satisfy the interest parity condition:

$$(1 + i_t) = (1 + i_t^*) \left(\frac{E_t}{E_{t+1}^e} \right) \quad (13)$$

4. Now suppose a country pegs the exchange rate at $E_t = \bar{E}$, so the current exchange rate $E_t = \bar{E}$. If financial and foreign exchange markets believe that the exchange rate will remain pegged at this value, then their expectation of the future exchange rate, E_{t+1}^e , is also equal to $\bar{E}$, and the interest parity relation becomes:

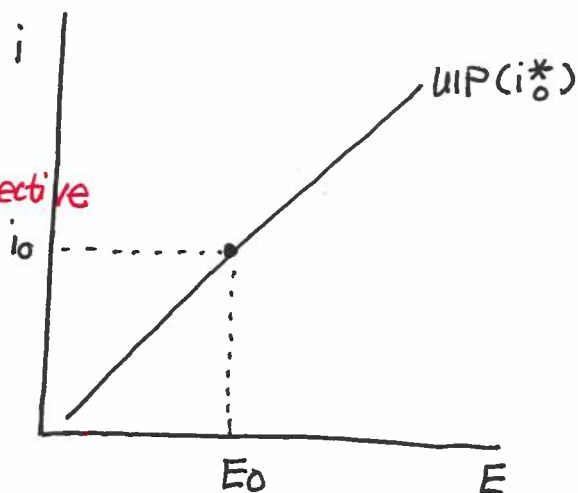
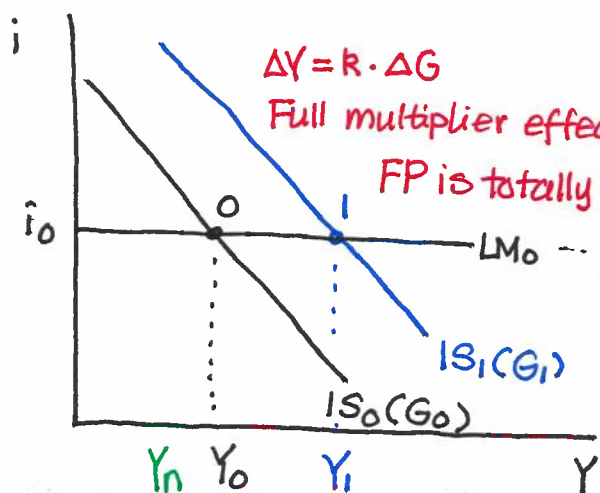
$$(1 + i_t) = (1 + i_t^*) \quad (14)$$

This means that:

$$i_t = i_t^* \quad (15)$$

5. In words: if financial investors expect the exchange rate to remain unchanged, they will require the same nominal interest rate in both countries. Under a fixed exchange rate and perfect capital mobility, the domestic interest rate must be equal to the foreign interest rate.
6. Under fixed exchange rates, the central bank gives up monetary policy as a policy instrument. With a fixed exchange rate, the domestic interest rate must be equal to the foreign interest rate.

Fixed ER + PCM + FP



Initially ΔY

Suppose now expansionary FP $\rightarrow G \uparrow (G_0 \rightarrow G_1) \rightarrow \vec{IS} (IS_0 \rightarrow IS_1)$

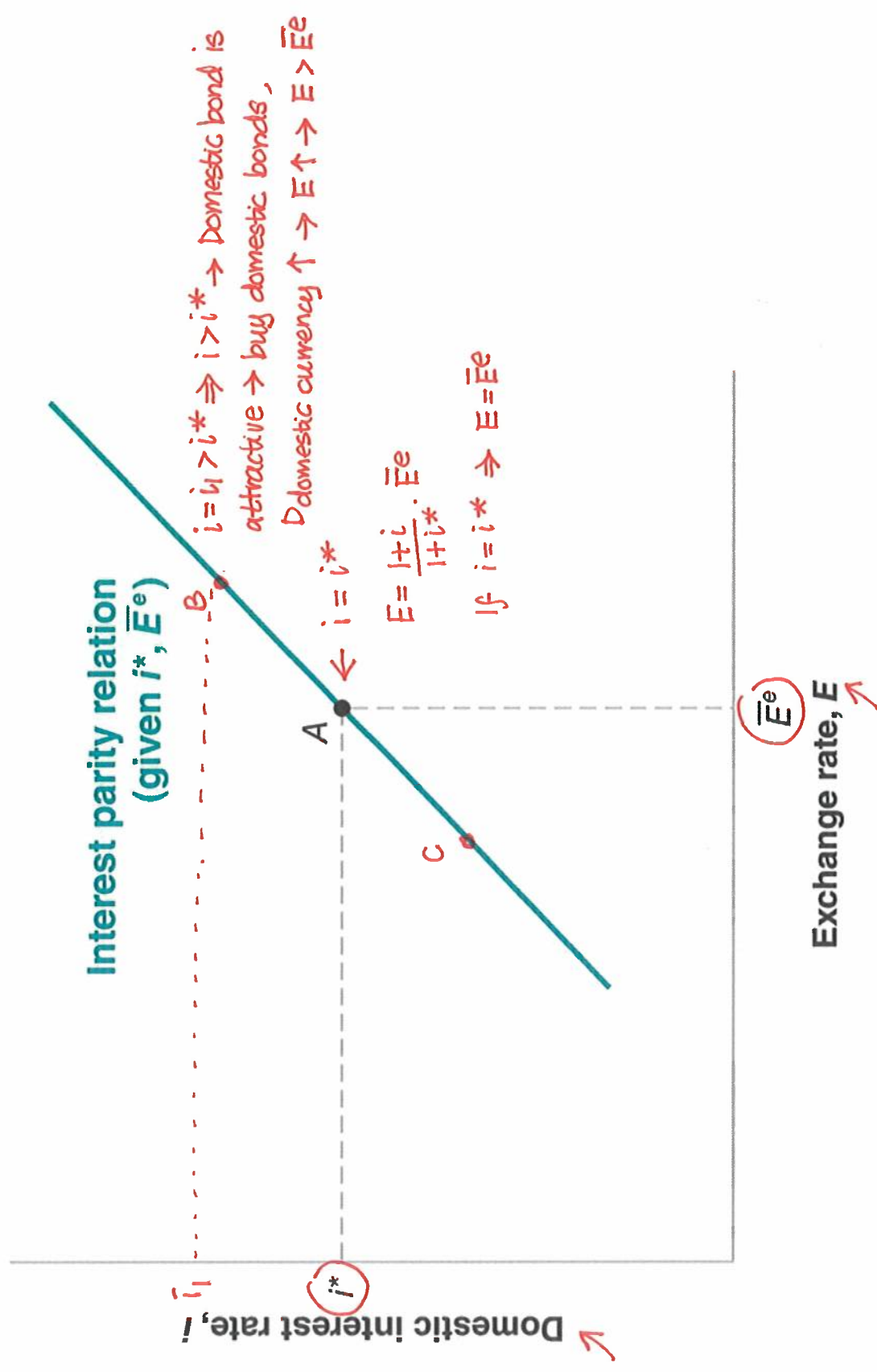
B/c $i = i_0 = i_0^* = i^*$, then $Y \uparrow (Y_0 \rightarrow Y_1)$

$E = E_0$

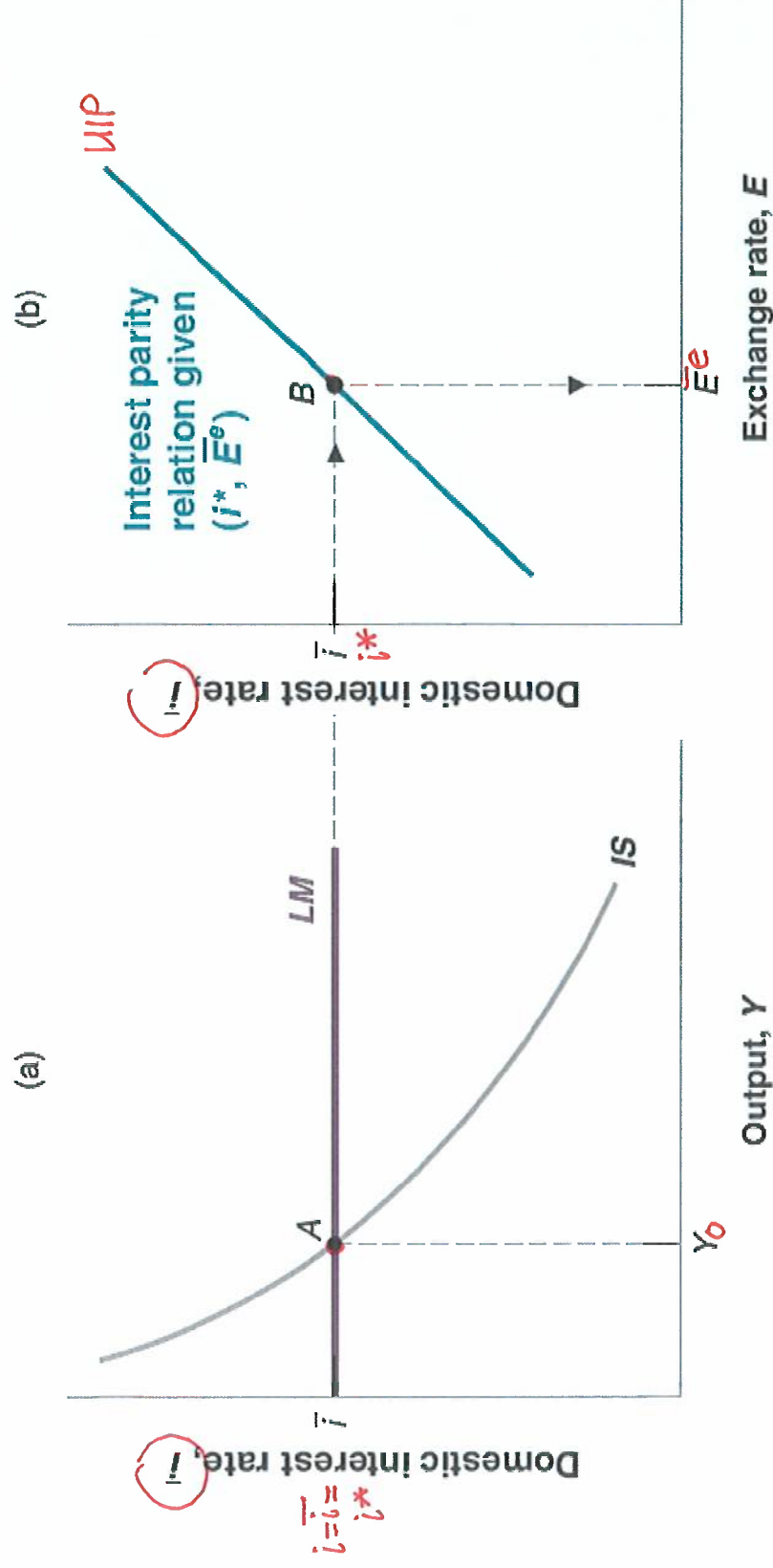
2 Fiscal Policy When the Exchange Rate is Fixed

1. If monetary policy can no longer be used under fixed exchange rates, what about fiscal policy?
2. The effects of an increase in government spending when the central bank pegs the exchange rate are identical to those for the case of flexible exchange rates and unchanged monetary policy. $i = i_0$
3. This is because, under flexible exchange rates, if the increase in spending, is not accompanied by a change in the interest rate, the exchange rate does not move. $\bar{E}$
4. Thus, when government spending increases, it makes no difference whether the country pegs its exchange rate.
5. The difference between fixed and flexible exchange rate is the ability of the central bank to respond.
*Fixed $\Rightarrow$ CB cannot respond by changing i
Flexible $\Rightarrow$ CB can respond*
6. If the increase in government spending pushed the economy above potential output, thus raising the possibility that inflation might increase, the central bank could respond by raising the interest rate. This option is no longer available under fixed exchange rates because the interest rate must be equal to the foreign rate. $Y > Y_n \rightarrow \text{Inflation}$
flexible
7. Then, why would a country choose to fix its exchange rate? You have seen a number of reasons why this appears to be a bad idea:
 - (a) By fixing the exchange rate, a country gives up a powerful tool for correcting trade imbalances or changing the level of economic activity.
 - (b) By committing to a particular exchange rate, a country also gives up control of its interest rate. Not only that, but the country must match movements in the foreign interest rate, at the risk of unwanted effects on its own activity.
 - (c) Although the country retains control of fiscal policy, one policy instrument may not be enough. For example, a fiscal expansion can help the economy get out of a recession, but only at the cost of a larger trade deficit. And a country that wants, for example, to decrease its budget deficit cannot, under fixed exchange rates, use monetary policy to offset the contractionary effect of its fiscal policy on output.
8. Why, then, do some countries fix their exchange rate?
9. Why have European countries adopted a common currency?

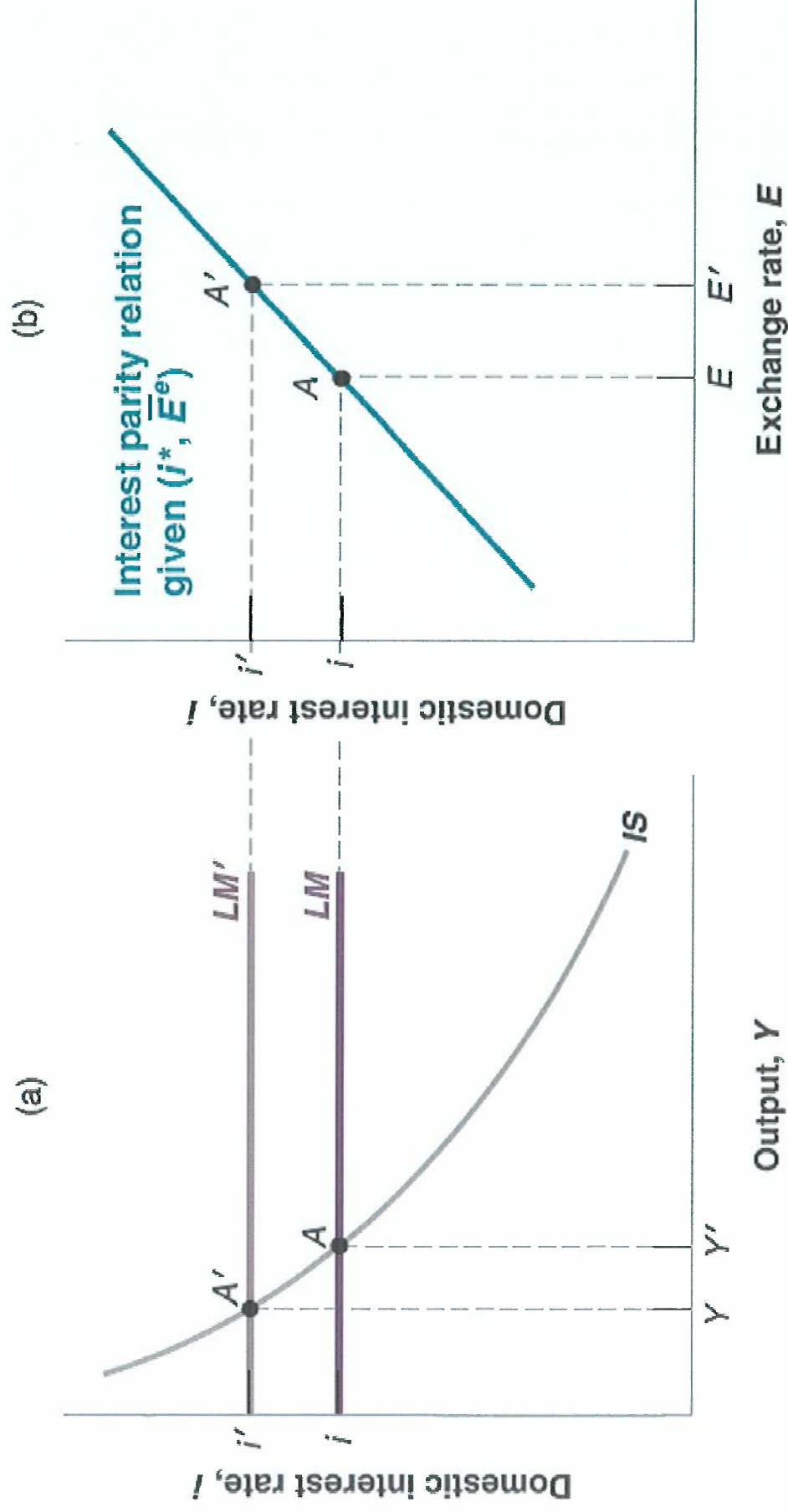
Equilibrium in Financial Markets



Putting Goods and Financial Markets Together



The Effects of Monetary Policy in an Open Economy



NANYANG TECHNOLOGICAL UNIVERSITY
School of Social Sciences
HE3002 Macroeconomics III

LECTURE 5
Exchange Rate Regimes

READINGS

1. Olivier Blanchard *Macroeconomics*, 8th Edition (2021), Prentice Hall, Chapter 20.

THE MEDIUM RUN $P \& P^*$ adjust

$Y < Y_n$ or $NX < 0$ MP (flexible)

1. Under flexible exchange rates, a country seeking real depreciation to reduce its trade deficit or recover from a recession could achieve this through expansionary monetary policy, leading to lower interest rates and a depreciated exchange rate.
P64
2. Under fixed exchange rates, a country lost both instruments: the nominal exchange rate couldn't be adjusted, and the UIP condition required the domestic interest rate to match the foreign rate, preventing interest rate adjustments. $E = \bar{E}$
 $i = i^*$
3. This makes a flexible exchange rate regime more attractive than a fixed exchange rate regime.
4. Why should a country give up two macroeconomic instruments?
5. In the medium run, when prices adjust, the difference between the two regimes fades away.
 $P \& P^*$ can adjust $\Rightarrow$ fixed & flexible are the same
6. In the medium run, the economy reaches the same real exchange rate and the same level of output, whether it operates under fixed or flexible exchange rates.
 ϵ Y
7. Recall the definition of the real exchange rate:

$$\epsilon = \frac{EP}{P^*} \quad (1)$$

$E \uparrow \rightarrow \text{app} \rightarrow \epsilon \uparrow$
 $P \uparrow \rightarrow \epsilon \uparrow$
 $P^* \downarrow \rightarrow \epsilon \uparrow$

8. There are two ways in which the real exchange rate can adjust:
 - (a) Through a change in nominal exchange rate, E . By definition, this can be done only under flexible exchange rates. And if we assume that P , and P^* , do not change in the short run, the only way to adjust the real exchange rate in the short run is through change in E .
 - (b) Through a change in P , relative to P^* in the medium run. This option is available even to a country operating under a fixed (nominal) exchange rate regime. The adjustment takes place through the price level rather than through the nominal exchange rate.

The IS Relation under Fixed Exchange Rates

1. In an open economy with fixed exchange rates, we can write the IS relation as:

$$Y = Y \left(\frac{\bar{E}P}{P^*}, G, T, i^* - \pi^e, Y^* \right)$$

$\begin{matrix} - & + & - & + \\ \bar{E}P & P^* & G & T & i^* - \pi^e & Y^* \end{matrix}$
 $\uparrow \quad \quad \quad \leftarrow \text{fixed}$

This is because

- (a) The real interest rate, r is equal to the nominal interest rate, i minus expected inflation, π^e :

$$r = i - \pi^e$$

- (b) The real exchange rate, ϵ

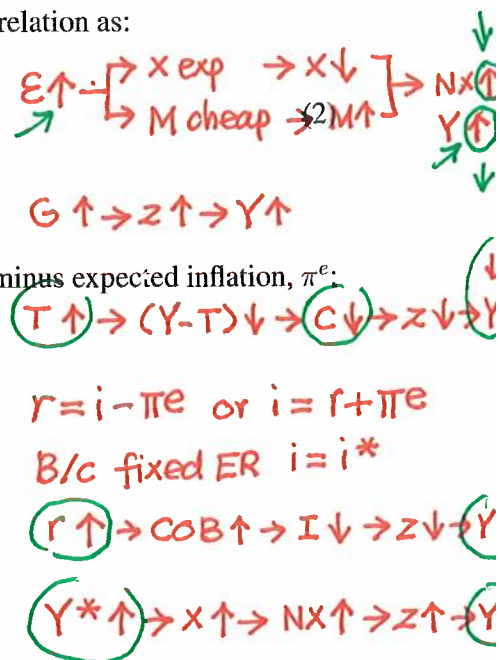
$$\epsilon = \frac{EP}{P^*}$$

- (c) Under fixed exchange rate,

$$E = \bar{E}$$

- (d) Under fixed exchange rates and perfect capital mobility,

$$i = i^*$$



2. Demand, and hence output, depend:

- negatively on the real exchange rate.
- positively on government spending and negatively on taxes.
- negatively on the domestic real interest rate.
- positively on foreign output through the effect on exports.

Equilibrium in the Short and the Medium Run

- Consider an economy where the real exchange rate is too high. As a result, the trade balance is in deficit, $NX < 0$ and output is below potential $Y < Y_n$.
- Under a flexible exchange rate regime,
 - $NX \uparrow \leftarrow E \text{ dep}$
 - (a) The central bank could decrease the interest rate, induce a nominal depreciation.
 - (b) Given the domestic and foreign price levels, which is fixed in the short run, the nominal depreciation would imply a real depreciation, an improve in trade balance and an increase in output. $E \text{ dep} \rightarrow \epsilon = \frac{E\bar{P}}{P^*}, \epsilon \text{ dep} \rightarrow NX \uparrow \rightarrow Z \uparrow \rightarrow Y \uparrow$
- Under a fixed exchange rate regime,

i is fixed $i = i^* \Rightarrow \bar{E} \Rightarrow$ No nominal dep $\left\{ \begin{array}{l} \rightarrow NX < 0 \\ \rightarrow Y < Y_n \end{array} \right\}$ SR

(a) The central bank cannot move the domestic interest rate. Thus, in the short run, the trade deficit remains, and the country remains in recession.

(b) In the medium run, MR then P & P^* can adjust. Note: ① IS $\rightarrow P \& P^*$ fixed SR
② PC $\rightarrow P \& P^*$ flexible M

(c) The Phillips curve relation:

HE2002

$$\pi - \pi^e = (\alpha/L)(Y - Y_n)$$

$Y > Y_n$ then $\pi > \pi^e$ (3)

$Y < Y_n$ then $\pi < \pi^e \Rightarrow \pi < \bar{\pi}$

$Y = Y_n$ then $\pi = \pi^e$

When $Y > Y_n$, then $\pi > \pi^e$. When $Y < Y_n$, then $\pi < \pi^e$.

(d) Suppose if expected inflation is assumed to be constant, then $\pi^e = \bar{\pi}$. This means that:

$Y = Y_n$ & $Y^* = Y_n^*$

$$\pi - \bar{\pi} = (\alpha/L)(Y - Y_n)$$

$\pi = \bar{\pi}$ when $Y = Y_n$ (4)

$$\pi = \bar{\pi} = \pi^*$$

$$\pi^* - \bar{\pi}^* = \left(\frac{\alpha}{L}\right)(Y^* - Y_n^*)$$

(e) Suppose that if both economies are at potential, inflation rates were the same, relative price levels would remain constant, and so would the real exchange rate. i.e. $\pi = \pi^* = \bar{\pi}$.

(f) As we assume that $Y < Y_n$, equation (4) implies that domestic inflation is lower than it would be if $Y = Y_n$, and thus lower than foreign inflation. $\pi < \bar{\pi}$ and $\pi < \pi^*$

(g) This means that the domestic price level increases more slowly than the foreign price level. Given the nominal exchange rate, which is fixed, the real exchange rate decreases.

(h) As a result, net exports increases over time, and so does output.

$$\varepsilon = \frac{EP}{P^*}$$

4. To summarize,

$$\bar{E} \Rightarrow \bar{\varepsilon}$$

(a) In the short run, a fixed nominal exchange rate implies a fixed real exchange rate.

(b) In the medium run, the real exchange rate can adjust even if nominal exchange rate is fixed.

This adjustment is achieved through movements in relative price levels over time.

$$\bar{E} \rightarrow \varepsilon \downarrow (\text{dep}^n) \rightarrow NX \uparrow$$

$$\ln \varepsilon = \ln E + \ln P - \ln P^*$$

$$\% \Delta \ln \varepsilon = \% \Delta \ln E + \pi - \pi^*$$

$$\pi < \pi^*$$

$$\% \Delta \ln \varepsilon = \pi - \pi^* \text{ b/c}$$

$$\% \Delta \ln E = 0$$

EXCHANGE RATE CRISES UNDER FIXED EXCHANGE RATES

$$\pi < \pi^*$$

1. Suppose a country has chosen to operate under a fixed exchange rate. $\% \Delta \ln \varepsilon < 0$ $\varepsilon \downarrow$ real dep.

2. Suppose financial investors begin to anticipate an exchange rate adjustment, either through devaluation or a shift to a flexible exchange rate regime with depreciation.

(a) The real exchange rate may be too high. Or, the domestic currency may be over-valued. In this case, a real depreciation is called for. Although this could be achieved in the medium run without a devaluation, financial investors might conclude that the government will take the quickest way out and devalue. Such an overvaluation is likely to happen in a country that pegs its nominal exchange rate to the currency of a country with lower inflation. Higher relative inflation implies a steadily increasing price of domestic goods relative to foreign

goods, a steady real appreciation, and so a steady worsening of the trade position. As time passes, the need for an adjustment of the real exchange rate increases, and financial investors become more and more nervous. They start thinking that a devaluation might be coming.

if $i \downarrow \rightarrow$ UIP suggests $E \downarrow$ (Not available under fixed)
 (b) Internal conditions may demand a lower domestic interest rate, which is not possible under fixed exchange rates. However, it can be achieved by switching to a flexible exchange rate regime. If the country allows the exchange rate to float and lowers its domestic interest rate, this will lead to a nominal depreciation.

3. As soon as financial markets believe a devaluation may be coming, maintaining the exchange rate requires an increase, often a large one, in the domestic interest rate.
4. To see this, return to the interest parity condition:

$$i_t = i_t^* - \frac{(E_{t+1}^e - E_t)}{E_t} \quad (5)$$

5. We interpreted this equation as a relation between the one-year domestic and foreign nominal interest rates, the current exchange rate and the expected exchange rate a year. But the choice of one year as the period was arbitrary.
6. The relation holds over short periods such as a day, week, or month. If financial markets expect the exchange rate to drop by 2% in a month, they will only hold domestic bonds if the one-month domestic interest rate exceeds the foreign interest rate by 2%. Expressed annually, this means the domestic interest rate must be 24% higher than the foreign rate ($2\% \times 12 = 24\%$).
7. Under fixed exchange rates, the current exchange rate, E_t , is set at some level, say $E_t = \bar{E}$.
8. If markets expect that the parity will be maintained over the period, then $E_{t+1}^e = \bar{E}$, and the UIP condition simply states that the domestic and the foreign interest rates must be equal.
9. Suppose, however, that participants in financial markets start anticipating a devaluation, a decrease in the exchange rate.

10. Suppose they believe that, over the coming month, there is a 75% chance the parity will be maintained and a 25% chance there will be a 20% devaluation.

11. The term $(E_{t+1}^e - E_t)/E_t$ which we assumed equal to 0 earlier, now equals:

$$0.75 \times 0\% + 0.25 \times 20\% = 5\%$$

12. That is, a 75% chance of no change plus a 25% chance of a devaluation of 20%.

Uncertain CB vs financial investors
 0.75 UIP hold CB 0%
 0.25 UIP cannot hold devaluation 20%

Thai Baht is overvalued.

Expect Baht to devalue (fixed ER).

Domestic bonds are not attractive & are risky.

To convince financial investors to hold domestic bonds $\rightarrow$
 $i \uparrow$

B/c $i \uparrow$ can prevent FO

But $i \uparrow \rightarrow r \uparrow \rightarrow COB \uparrow \rightarrow I \downarrow$

If T , then FO happens $\rightarrow E$ has a tend to dep

To prevent $E \downarrow \rightarrow$ FX Intervention $\left\{ \begin{array}{l} \rightarrow \text{buy Thai baht} \rightarrow Ms \downarrow \rightarrow i \uparrow \\ \rightarrow \text{sell FR} \rightarrow FR \downarrow \end{array} \right.$

13. This implies that, if the central bank wants to maintain the existing parity, it must now set a **monthly** interest rate 5% higher than before – 60% higher **annually** (12 months $\times$ 5% per month) is the interest differential needed to convince investors to hold domestic bonds rather than foreign bonds.
14. Any smaller interest differential, and investors will not want to hold domestic bonds.
15. What choices, then, do the government and the central bank have?
- The government and central bank can attempt to reassure markets by asserting they have no plans to devalue. Prime ministers may appear on TV to reaffirm their strong commitment to maintaining the current exchange rate parity.
 - The central bank may raise interest rates, but not sufficiently to offset the perceived risk of devaluation. While domestic interest rates are high, they still fail to fully compensate investors for the risk. As a result, large financial outflows occur, as investors shift from domestic to foreign bonds. They sell domestic bonds for domestic currency, then exchange it for foreign currency to purchase foreign bonds.
 - No intervention:** The large sales of domestic currency for foreign currency would lead to a depreciation.
 - With intervention:** The central bank must therefore stand ready to buy domestic currency and sell foreign currency at the current exchange rate. In doing so, it often loses most of its reserves of foreign currency. $i \uparrow \rightarrow i > i^* \rightarrow \text{prevent FO}$
However $i \uparrow \rightarrow r \uparrow \rightarrow \text{COB} \uparrow \rightarrow I \downarrow$
 - Eventually, the choice for the central bank becomes either to increase the interest rate enough to satisfy Equation (5), or to validate the market's expectations and devalue.
 - Setting a very high short-term domestic interest rate can have a devastating effect on demand and on output – no firm wants to invest and no consumer wants to borrow when interest rates are very high.
 - This course of action makes sense only if (1) the perceived probability of a devaluation is small, so the interest rate does not have to be too high, and (2) the government believes markets will soon become convinced that no devaluation is coming, allowing domestic interest rates to decrease.
 - Otherwise the only option is to devalue.
16. To summarise, expectations that a devaluation may be coming can trigger an exchange rate crisis. Faced with such expectations, the government has two options:
- Give in and devalue.
 - Fight and maintain the parity, at the cost of very high interest rates and a potential recession.

EXCHANGE RATE MOVEMENTS UNDER FLEXIBLE EXCHANGE RATE

1. In the model we developed, there was a simple relation between the interest rate and the exchange rate: the lower the interest rate, the lower the exchange rate.
2. This implied that a country that wanted to maintain a stable exchange rate just had to maintain its interest rate close to the foreign interest rate.
3. A country that wanted to achieve a given depreciation just had to decrease its interest rate by the right amount.
4. In reality, exchange rates often move even in the absence of movements in interest rates. Furthermore, the size of the effect of a given change in the interest rate on the exchange rate is hard to predict.
5. This makes it much harder for monetary policy to achieve its desired outcome.
6. To see why things are more complicated, we must return once again to the interest parity condition:

$$(1 + i_t) = (1 + i_t^*) \left(\frac{E_t}{E_{t+1}^e} \right) \quad (6)$$

7. Arranging this gives:

$$E_t = \frac{1 + i_t}{1 + i_t^*} E_{t+1}^e \quad (7)$$

8. Think of the time period (from t to $t + 1$) as one year.
9. The exchange rate this year depends on the one-year domestic interest rate, the one-year foreign interest rate and the exchange rate expected for next year.
10. We assumed that the expected exchange rate next year, E_{t+1}^e was constant, but that was a simplification.
11. The exchange rate expected one year hence is not constant.
12. Using Equation (7), but now for next year, we can write Equation (7) for year $t + 1$ as:

$$E_{t+1} = \frac{1 + i_{t+1}}{1 + i_{t+1}^*} E_{t+2}^e \quad (8)$$

13. The exchange rate in year $t + 1$ depends on the domestic interest rate and the foreign interest rate for year $t + 1$, as well as on the expected future exchange rate in year $t + 2$. So, the expectation of the exchange rate in year $t + 1$, held as of year t , is given by:

$$E_{t+1}^e = \frac{1 + i_{t+1}^e}{1 + i_{t+1}^{*e}} E_{t+2}^e \quad (9)$$

14. Replacing E_{t+1}^e with the expression above gives:

$$E_t = \frac{(1 + i_t)(1 + i_{t+1}^e)}{(1 + i_t^*)(1 + i_{t+1}^{*e})} E_{t+2}^e \quad (10)$$

15. The current exchange rate depends on this year's domestic and foreign interest rates, on next year's expected domestic and foreign interest rates, and on the expected exchange rate two years from now. Continuing to solve forward in time in the same way we get:

$$E_t = \frac{(1 + i_t)(1 + i_{t+1}^e) \dots (1 + i_{t+n}^e)}{(1 + i_t^*)(1 + i_{t+1}^{*e}) \dots (1 + i_{t+n}^{*e})} E_{t+n+1}^e \quad (11)$$

16. Suppose we take n to be large, say 10 years, this relation tells us that the current exchange rate depends on two sets of factors:

- (a) Current and expected domestic and foreign interest rates for each year over the next ten years.
- (b) The expected exchange rate 10 years from now.

17. Equation (11) makes the three points we want to make here:

- (a) The level of today's exchange rate will move one-for-one with the future expected exchange rate.
- (b) Today's exchange rate will move when future expected interest rates move in either country.
- (c) Because today's exchange rate moves with any change in expectations, the exchange rate will be volatile.